MATH4511 Quantitative Methods for Fixed Income Securities
Problem Set 1

Introduction to interest rate derivatives (Part 1)

Problem 1

We consider a 6-year coupon bond with face value $1000 and annual coupon rate 6% payable quarterly. Suppose that the current annual effective yield rate of the bond is 5%

(a) Find the current price of the bond.

(b) Suppose that the yield rate drops to 4.6% after 2.5 years (30 months), calculate the dirty price of the bond at that time. How about the clean price of this bond?

Problem 2

We consider a n-year bond (where $n \geq 4$ is a positive integer) with face value $\$100$ and annual coupon rate $c = 5\%$ payable annually. The annual nominal yield rate of the bond is currently 6%. Suppose that

- The *clean price* of the bond after 15 months is 95.94321.
- The yield rate of the bond remains unchanged during first 3 years.

(a) Find the bond price after 3 years.

(b) Find the bond price (quoted as clean price) of the bond after 25 months.

Problem 3

We consider a 8-year coupon bond with face value $800. The bond pays coupon semi-annually at an annual coupon rate $c = 4\%$. The annual effective yield rate of the bond is currently $i = 5\%$.

Suppose that the bond price after 3 years is *lower than* the current price, what can you say about the yield rate of the bond at that time? (Choose one of the following)

A. Lower than 5%

B. Equal 5%

C. Higher than 5%

(☺Hint: You can first consider the case if the yield rate remains to be 5% after 3 years)

Problem 4

The following table shows the current spot rates (quoted as annual effective interest ate) of various maturities:

	3 months	6 months	12 months	18 months	24 months
Spot rate	1.5%	1.6%	2%	2.2%	2.3%

(a) We consider a two-year bond with face value $1000 and annual coupon rate $c = 6\%$ payable annually. Find the annual effective yield rate of the bond.

(b) As mentioned in the lecture, it is possible that the spot rate can be quoted as nominal interest rate. Determine the corresponding set of spot rates (3 months, 6 months, 12 months, 18 months and 24 months) if the spot rates are quoted as

(i) annual nominal interest rate compounded continuously.

(ii) annual nominal interest rate compounded semi-annually.

Problem 5

We adopt the same setup as in Problem 4 and consider a 2-year bond with face value \$500 and annual coupon rate $c = 3.6\%$ payable semi-annually. We let P_0 and P_1 be the current bond price and bond price after 1 year respectively.

Suppose that the spot rates are increased uniformly by 0.2% after 1 year, find $P_1 - P_0$.

Problem 6 (Obtaining term structure of interest rate using bond prices)

(a) The following table shows the market prices of some coupon bonds

	Bond term	Face value	Annual coupon rate	Coupon frequency	Market price
Bond A	1 year	1000	Nil	Nil	970.8738
Bond B	2 years	1200	3%	Annual	1200
Bond C	3 years	1200	5%	Annual	1264.5347
Bond D	4 years	1400	3%	Annual	1426.8603

(*Note: Bond A is a zero-coupon bond)

Determine the 1-year spot rate, 2-year spot rate, 3-year spot rate and 4-year spot rate. Quote the spot rate as annual effective interest/yield rate.

(b) The following tables shows the information of three bonds available in the market:

	Bond term	Face value	Annual coupon rate	Coupon frequency	Yield rate (*)
Bond A	1 year	1000	Nil	Nil	4.6%
Bond B	2 years	1000	6%	Annual	4.9%
Bond C	3 years	1000	Nil	Nil	5.5%

(* The yield rate is quoted as annual nominal yield rate compounded continuously)

Determine the 1-year spot rate, 2-year spot rate and 3-year spot rate. Quote the spot rate as annual effective interest/yield rate.

Problem 7 (Problem 10 of MATH2511 2017 Final)

The following table shows the partial information of the current spot rates of different maturities (quoted as annual effective interest rate):

Term t (years)	0.5	1	1.5	2	2.5	3	⋯	10
Spot rate $s_0(t)$	2.5%	??	??	??	??	??	⋯	??

You are given that

- The current term structure of interest rate is either normal, inverted or flat.
- The current annual nominal yield rate of a 10-year straight-term bond is 2.49%. This bond has face value F and redemption value C. It pays coupon semi-annually.

What can you say about the term structure of the interest rate currently?

(A) Inverted
(B) Flat
(C) Normal

Choose the best answer and explain your answer.

(☺Hint: Firstly, calculate the annual effective yield rate of the bond)

Problem 8 (Harder)

You are given that

- The current price of 10-year coupon bond with face value \$100 and annual coupon rate 8% payable annually is \$105.
- The current price of 10-year coupon bond with face value \$100 and annual coupon rate 4% payable annually is \$95.

(a) Using no arbitrage pricing principle, find the current price of a 10-year zero coupon bond with face value \$200.

(b) Using no arbitrage pricing principle, find the current price of a 10-year coupon bond with face value \$500 and annual coupon rate 6%.

Problem 9

Compute the modified durations (with respect to annual effective yield rate) for the following bonds:

(a) An 8-year bond with face value \$100 and annual coupon rate 8% payable semi-annually. The annual effective yield rate of the bond is 7%.

(b) A 10-year bond with face value \$100 and annual coupon rate 6% payable annually. The current price of the bond is \$92.
(☺Hint: You need to calculate the annual effective yield rate of the bond first)

Problem 10

We consider a 12-year bond with face value \$1200 and annual coupon rate 4% payable semi-annually. You are given that the annual nominal yield rate of the bond is 3.6%. Using second-order approximation, estimate the bond price when the annual nominal yield rate is increased to 4.4%.

Problem 11

We consider a 12-year bond that pays coupon annually. The face value of the bond is \$4800 and the annual effective yield rate of the bond is currently 8.4%. You are given that

- If the annual effective yield rate is decreased by 0.2%, the percentage change in bond price, according to *first-order approximation*, is estimated to be 1.5086%.

Using second-order approximation, estimate the percentage change of the bond price if the annual nominal yield rate is increased by 0.15%.

Problem 12

We consider a 6-year floating rate note issued today. The face value of the bond is 100 and it pays coupon semi-annually. The coupon rate over a coupon payment is $\{6\ month\ spot\ rate\} + 0.15\%$ (or $R_{\left[\frac{i-1}{2},\frac{i}{2}\right]} + 0.0015\%$). The current 6-month spot rate is known to be 2% (quoted as annual effective rate).

(a) Calculate the amount of first coupon payment.

(b) Suppose that the current term structure is flat, calculate the current price of the bond.

(c) Suppose that the current term structure remains to be flat after 40 months and the spot rate is 2.6%. You are also given that the 6-month spot rate at previous payment date is 2.5%. Calculate the price of the bond at that time.

Problem 13

The following table summarizes the 1-year forward rates with various starting dates:

	Now	1 year	2 year	3 year	4 year
1-year forward rate	0.026	0.028001	0.028501	0.033513	0.039039

(*All rates are quoted as annual effective interest rate)

(a) Find the term structure of the interest rate by finding 1 year spot rate, 2 year spot rate, 3 year spot rate, 4 year spot rate and 5 year spot rate.

(b) We consider a 4-year coupon with face value \$1000 and annual coupon rate $c = 4\%$ payable annually. Find the current price of bond using

(i) Spot rate

(ii) Forward rate

Problem 14 (Harder)

Suppose that the current term structure is normal (upward-sloping). Rank the following in order of magnitude (from largest to smallest)

(1) The 5-year spot rate
(2) The yield rate on a 5-year coupon bond.
(3) The forward rate applied over the period $[4.75, 5]$. That is, $f_{[4.75,5]}$.

How about the case when the current term structure is flat or inverted?

Problem 15

Recently, an investor has entered into a 6-months forward contract (i.e. the contract will expire after 6 months) on a non-dividend paying asset when the asset price is \$30 and the 6-month spot rate is 4% (which is quoted as an annual effective yield rate).

(a) What is the delivery price (or 6-month forward price) of the contract?

(b) Suppose that the stock price moves up to \$35 after 3 months, what is the value of this forward contract at that time? How about the case when the stock price moves down to \$25 after 3 months? Assume that the 3-month spot rate at that time is 2% (quoted as annual effective yield rate).

Problem 16

The following table shows the spot rates of various maturities. Here, all rates are quoted as annual effective interest rate.

	6 months	12 months	18 months	24 months
Spot rates	4%	4.5%	4.75%	5%

(a) Determine the 6-month forward rate, starting in 18 months. That is, $f_{[1.5,2]}$.

(b) We consider a forward rate agreement which the investor (floating rate receiver) can exchange fixed rate payment for floating rate payment over 6 months, starting in 18 months. The notional principal of the FRA is \$10000. The fixed rate is set to be 6% and the floating rate is set at $L_{[1.5,2]}$. Both rates are quoted as annual effective interest rate.
Compute the current value of the FRA.

Problem 17 (Harder)

(a) We consider a T-year forward contract on an asset. It is known that the asset will pay a single dividend of amount D at time t_D, where $0 < t_D < T$. We let S_0 be the current price of the asset. Using no arbitrage pricing principle, show that the delivery price (i.e. T-year forward price) of the contract is

$$K = f_{[0,T]} = \frac{S_0 - DB(0; t_D)}{B(0; T)},$$

where $B(0; T)$ denote the current price of T-year zero-coupon bond with face value 1.

(b) For any $0 < t < T$, find the value of the forward contract initialized at time 0. Express your answer in terms of $K, S_t, B(0; ??)$ and D.

Problem 18

You are given that

- The current 6-month spot rate and 9-month spot rate are 5% and 6% respectively.
- The market 3-month forward rate, starting in 6 months, is $f_{[0.5,0.75]} = 7\%$.
- All rates are quoted as annual effective interest rate.

Identify an arbitrage opportunity. (☺Hint: Consider FRA)

Problem 19

The current price of a non-divided paying stock is $55. The current 3-month spot rate and 6-month spot rate are 2.5% and 3% respectively. Here, the two rates are quoted as annual effective interest rate.

(a) If 6-month forward price is $56, identify an arbitrage opportunity.

(b) If 3-month forward price is $56, identify an arbitrage opportunity.

Problem 20

The current exchange rate between the Swiss Franc and US dollar is 1 Franc = USD$ 1.0404. You are given that the current 3-month interest rate (spot rate) of USD currency and Swiss Franc currency are 1% and 0.5% respectively. Here, the spot rates are quoted as annual nominal interest rate convertible continuously. Suppose that the 3-month forward exchange rate is 1 Franc = USD$ 1.03. Identify an arbitrage opportunity.

Problem 21

We consider a 2-year forward contract on a 10-year coupon bond. The bond has face value $F = 1000$ and annual coupon rate $c = 6\%$ payable annually. The annual effective yield rate of the bond is 5%. You are also given that 1-year spot rate and 2-year spot rate, quoted as annual effective yield rate, are 4.5% and 4.9% respectively.

(a) Compute the delivery price of the forward contract.
(☺Hint: Compute the bond price first.)

(b) After 1 year, the yield rate of the bond decreased to 4.8%. The 1-year spot rate and 2-year spot rate becomes 4.6% and 4.8% respectively. Find the value of the forward contract.
(☺Hint: Be careful that one coupon has been paid and the bond has remaining life 9 years).

Problem 22 (Harder)

(a) We consider a T-year forward contract on an asset.. Suppose that the underlying asset pays dividends $D_1, D_2, \dots, D_n$ at times $t_1, t_2, \dots, t_n$ respectively, where $0 < t_1 < t_2 < \cdots < t_n \leq T$. In other words, this asset pays n dividends over the first T years.

Using no arbitrage pricing principle, show that the delivery price (T-year forward price) is given by

$$K = f_{[0,T]} = \frac{S_0}{B(0;T)} - \sum_{i=1}^{n} \frac{D_i B(0;t_i)}{B(0;T)}.$$

(b) The current price of an asset is S_0. You are given that

- The asset pays two dividends of amount D each after 4 months and after 8 months respectively.
- The 3-month forward price on this asset is \$42.8.
- The 9-month forward price on this asset is \$42.003.
- The current term structure is flat and the annual effective interest rate is 6.822672%

Find the 6-month forward price (i.e. $f_{[0,0.5]}$) on this asset.